Question	Cell organelles present in
Type	multiple choice
Option	some cell
Option	all cells
Option	bacterial cell only
Option	virus and prions
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Histones are present in …… structure of Eukaryotes
Type	multiple choice
Option	DNA
Option	Chromatin
Option	Chromosome
Option	both (b) and (c)
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is an example of haploid
Type	multiple choice
Option	Spore and gamete
Option	Egg and zygote
Option	Embryo and sperm
Option	Egg and somatic cell in diploid animal
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Genetic material is DNA in
Type	multiple choice
Option	*E. coli* and viroids
Option	*Streptococcus*, *Nostoc* and *E. coli*
Option	Bacteriophage and TMV
Option	Yeast and HIV
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Not common in prokaryotes and eukaryotes
Type	multiple choice
Option	Cell membrane
Option	Ribosome in cytoplasm
Option	Location of DNA
Option	Genetic material is DNA
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	DNA replication absent in
Type	multiple choice
Option	*Nostoc*

Option	Yeast
Option	Human cell
Option	Neutron
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Organelles present in both *E. coli* and Amoeba
Type	multiple choice
Option	Nucleus and mitochondria
Option	Ribosome
Option	ER and Golgi bodies
Option	All
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Gap B/w DNA replication and cell division is
Type	multiple choice
Option	G_0
Option	G_2
Option	M
Option	G_1
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Interphase does not involve
Type	multiple choice
Option	G_1
Option	S-phase
Option	G_2-phase
Option	M-phase
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	DNA replication in yeast occurs in
Type	multiple choice
Option	S-phase
Option	G_1-phase
Option	Interphase
Option	G_2-phase
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Mark the incorrect statement
Type	multiple choice
Option	G_1 phase shows growth
Option	G_2 phase also shows growth
Option	DNA replication occurs in M-phase
Option	Centriole present in animal cell

Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Number of chromosome in G_2 phase in animal cell
Type	multiple choice
Option	Two
Option	One
Option	Three
Option	Four
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Chromosomes are visible in nucleus of cell during which phase of cell cycle
Type	multiple choice
Option	Interphase
Option	M-phase
Option	G_1-phase
Option	G_2-phase
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following event occurs in S-phase
Type	multiple choice
Option	Centriole doubling inside nucleus
Option	Centriole doubling in cytoplasm of plant cell
Option	Centriole doubling in cytoplasm of animal cell
Option	DNA amount become half
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Consider the following and mark the incorrectly matched
Type	multiple choice
Option	G_1 phase – Cell growth
Option	G_2 phase – Cell prepare for division
Option	S-phase – DNA replication in nucleus
Option	M-phase – Centriole duplication
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following does not involve S-phase
Type	multiple choice
Option	Meiosis
Option	Binary fission in bacteria
Option	Mitosis
Option	All
Answer	2
Solution	
Positive Marks	4

Negative Marks	1
Question	Number of chromosomes in human diploid cell in G_2 phase
Type	multiple choice
Option	46
Option	92
Option	43
Option	23
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Mitosis occurs generally in
Type	multiple choice
Option	Diploid animal cell
Option	Haploid plant cell
Option	Haploid animal cell
Option	All diploid cell
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Which is not possible
Type	multiple choice
Option	Human cell with 46 chromosome in G_1
Option	Human cell with 23 chromosome in G_2
Option	Human cell with 46 chromosome in G_2
Option	Human cell with 23 chromosome in G_1
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Diploid cell (2n = 24), number of chromosome in egg cell
Type	multiple choice
Option	48
Option	24
Option	12
Option	6
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In a pollen grain if chromosome number is 17, number of chromosome in diploid cell in G_2
Type	multiple choice
Option	68
Option	34
Option	17
Option	None
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	M-phase is known as equational division as

Type	multiple choice
Option	Since number of chromosome in parent and progeny cells is not same
Option	Since number of chromatids in parent and progeny cells is not same
Option	Since number of chromatids in parent and progeny cells is the same
Option	Since number of chromosomes in parent and progeny cells is the same
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The new DNA molecules formed in S-phase are not distinct but intertwined in
Type	multiple choice
Option	Prophase
Option	Metaphase
Option	Anaphase
Option	G_2-phase
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Change which does not occur in metaphase of mitosis
Type	multiple choice
Option	Condensation of chromosome material is completed
Option	Clearly visible chromosomes in microscope
Option	Separation of sister chromatid
Option	Kinetochore attach to spindle
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In which phase chromosome with two chromatids first appear
Type	multiple choice
Option	Anaphase
Option	Metaphase
Option	Prophase
Option	Telophase
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In which phase two sister chromatids of one chromosome get separated
Type	multiple choice
Option	Anaphase
Option	Metaphase
Option	Prophase
Option	Telophase
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	In which phase chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements
Type	multiple choice
Option	Anaphase

Option	Metaphase
Option	Prophase
Option	Telophase
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Mark the incorrect
Type	multiple choice
Option	In anaphase and metaphase chromosomes are in cytoplasm
Option	Chromosomes are moved to spindle equator and get aligned along metaphase plate through spindle fibres
Option	In anaphase daughter chromosomes move away from equatorial plate
Option	Some organisms karyokinesis is not followed by cytokinesis resulting in formation of syncytium
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In _____ phase of mitosis chromosome are in cytoplasm with two chromatid attach on centromere
Type	multiple choice
Option	Anaphase and Telophase
Option	Metaphase and Prophase
Option	Prophase
Option	Metaphase
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In metaphase of mitosis the first event is-
Type	multiple choice
Option	Disintegration of nuclear envelop
Option	Chromosome condensation start
Option	Chromosome come at metaphasic plate
Option	Spindle fibre attach on chromosome
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Phase in which chromosomes are in cytoplasm
Type	multiple choice
Option	Prophase
Option	Metaphase
Option	Metaphase & Anaphase
Option	Telophase
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Chromosome in mitosis is fully condensed in
Type	multiple choice
Option	Metaphase

Option	Telophase
Option	Anaphase
Option	Prophase
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Chromosome with two sister chromatids first appear in
Type	multiple choice
Option	Prophase
Option	Anaphase
Option	Telophase
Option	Metaphase
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	In *Drosophila* (2n = 8), number of kinetochore in metaphase
Type	multiple choice
Option	16
Option	8
Option	32
Option	4
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Condensation of chromatin material completed in
Type	multiple choice
Option	Prophase
Option	Metaphase
Option	Anaphase
Option	Telophase
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Centripetal cytokinesis is
Type	multiple choice
Option	Occur in plant
Option	Cell plate method
Option	Cleavage method
Option	All
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In anaphase chromatids separate to form
Type	multiple choice
Option	Sister chromatid
Option	Daughter chromosome
Option	Half the number of chromosome
Option	Daughter chromatids

Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In which phase nuclear envelope develops around the chromosome clusters at each pole forming two daughter nuclei
Type	multiple choice
Option	Anaphase
Option	Metaphase
Option	Prophase
Option	Telophase
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Mark the correctly matched
Type	multiple choice
Option	Anaphase – Separation of non-sister chromatids
Option	Metaphase – Kinetochore disappears
Option	Telophase – Nuclear membrane reappears and Golgi disintegrate
Option	Prophase – Chromosome condensation not complete
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The stage at which morphology of chromosome is most easily studied is
Type	multiple choice
Option	Anaphase
Option	Metaphase
Option	Prophase
Option	Telophase
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	In anaphase each chromosome moves away from the equatorial plate, the centromere remains directed towards the pole and hence at the leading edge, with arms of the chromosome
Type	Move together with centromere
Option	Arms trailing behind
Option	Arms ahead of centromere
Option	Arms remain at equator
Option	2
Answer	
Solution	4
Positive Marks	1
Negative Marks	The centrosome, which had undergone duplication during S-phase of interphase, now begins to move towards opposite poles of the cells. This feature is of
Question	multiple choice
Type	Anaphase
Option	Metaphase
Option	Prophase
Option	Telophase

Option	3
Answer	
Solution	4
Positive Marks	1
Negative Marks	Number of chromosomal fibre or spindle fibre attached to chromosome in pea plant (2n = 14) undergoing mitosis is
Question	multiple choice
Type	14
Option	28
Option	7
Option	10
Option	2
Answer	
Solution	4
Positive Marks	1
Negative Marks	Mark the incorrectly matched
Question	multiple choice
Type	Mitosis – occur in diploid cell
Option	Mitosis – occur in haploid cell of social insects
Option	Meiosis – occur in diploid cell
Option	Meiosis – occur in haploid cell
Option	4
Answer	
Solution	
Positive Marks	4
Negative Marks	1
Question	Middle lamella is made up of
Type	multiple choice
Option	Calcium pectate
Option	Magnesium pectate
Option	Spindle fibre
Option	Cell wall
Answer	1
Solution	
Positive Marks	4
Negative Marks	1